

MATH 195: 9/17
RECIPROCAL OF POWER FUNCTIONS

The *reciprocal* of A is what you multiply A by to get 1, namely $\frac{1}{A}$. The reciprocals of power functions are those functions of the form $f(x) = x^{-n} = \frac{1}{x^n}$ for n a positive integer.

The general shape of the graph of x^{-n} depends on the *parity* of n .



Vertical asymptote at $x = 0$ and *horizontal asymptote* at $y = 0$. We write $x^{-n} \rightarrow \pm\infty$ as $x \rightarrow 0$ and $x^{-n} \rightarrow 0$ as $x \rightarrow \pm\infty$ to express the behavior of the vertical and horizontal asymptote. This notation can be used to also describe other kinds of end behavior.

- (1) Explain why these graphs have a vertical asymptote at $x = 0$. What is the domain of x^{-n} ?
- (2) Why does the parity determine whether x^{-n} goes off to ∞ or $-\infty$ as x approaches 0 from the left?
- (3) Explain why these graphs have a horizontal asymptote at $y = 0$. That is, what value does x^{-n} approach as $x \rightarrow \pm\infty$?
- (4) Sketch graphs of x^{-1} and x^{-2} on the domain $(0, \infty)$. What looks different about them?
- (5) More generally, what role does n play in determining the shape of the graph of x^{-n} ?

When sketching a graph of a geometric transformation of a reciprocal of a power function, there are two more important details: (1) Where are the asymptotes? (2) What is the orientation? The first is answered by identifying the horizontal and vertical translations. The second is answered by determining whether there was a vertical reflection.

- (1) Explain why $a(x) = \frac{1}{x-2} + 3$ has its asymptotes at $x = 2$ and $y = 3$. Sketch a graph, drawing in its asymptotes. What is its domain and range?
- (2) Sketch a graph of $b(x) = -2(x+1)^{-2} + 2$, drawing in its asymptotes. What are its domain? Range? Where is it increasing? Decreasing? Concave up? Concave down?
- (3) Determine a general rule for the domain and range of a reciprocal of a power function, knowing it has asymptotes at $x = h$ and $y = k$. [Hint: your answer should bifurcate based on the parity of n .]
- (4)
- (5) Explain why a horizontal reflection on a reciprocal of a power function of even parity doesn't change the graph. Give both a graphical and algebraic explanation.
- (6) Explain why a horizontal reflection on a reciprocal of a power function of odd parity is equivalent to doing a vertical reflection. Give both a graphical and algebraic explanation.
- (7) Explain why a horizontal scaling on a reciprocal of a power function is equivalent to a vertical scaling by a different amount. Determine the factor of the vertical scaling equivalent to a horizontal scaling by a . [Hint: your answer will depend upon n .]
- (8) Write, as simply as possible, a general form for a function which is a geometric transformation of a reciprocal of a power function. Why does your form cover all six geometric transformations?

Reciprocals of power functions are our first functions with asymptotes. We will later study other kinds of functions with asymptotes. Before we study them algebraically we can think about what is possible for graphs of functions.

- (1) Reciprocals of power functions always have the same horizontal asymptote on the left and right. Draw a graph of a function with two different horizontal asymptotes. Is it possible for a function to have three different horizontal asymptotes? Explain.
- (2) Draw a graph of a function with two different vertical asymptotes. Draw one with three different asymptotes. Can a function have 17 different vertical asymptotes? Can a function have infinitely many different vertical asymptotes? Explain.
- (3) Draw a graph of a function with one horizontal asymptote but no vertical asymptotes.
- (4) Draw a graph of a function with one vertical asymptote but no horizontal asymptotes.
- (5) Draw a graph of a function with a horizontal asymptote on the left but none on the right.
- (6) Draw a graph of a function with a vertical asymptote at $x = 0$ and domain $(0, \infty)$.
- (7) Draw a graph of a function $f(x)$ with no horizontal asymptote but $f(x) \not\rightarrow \pm\infty$ as $x \rightarrow \pm\infty$.